Increasing electricity exports through delivery-informed allocation in distribution networks

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SUMMARY

Export limits protect distribution networks, but allocating permission before output is known can restrict one generator while another leaves permission unused. The difficulty is identifying when reallocation would let more electricity reach the network. We introduce externality-conditioned counterfactual feedback (ECF) for request-based allocation. It tests alternative allocations for completed intervals using available-export estimates and alternating-current power-flow checks. It lowers later request ceilings only when reallocation would increase delivery elsewhere and in total. In controlled simulations across five allocation rules, moderate feedback without a shortfall charge delivers 24.8 kW more on average than proportional reductions of the same total request. In an 80-day Australian public-data simulation, ECF increases modeled export by 1.39 MWh (0.59%) over no feedback and outperforms random updates at a similar frequency. Stronger updating can reverse the gain. The results favor selective reductions in later requests that preserve export as output recovers, rather than minimizing unused permission.

KEYWORDS

distributed generation, electricity exports, distribution networks, capacity allocation, counterfactual feedback

BROADER CONTEXT

Distributed generation supplies electricity close to demand, but exporting the surplus requires shared network capacity. Flexible export arrangements allow that capacity to be used as conditions change. They also raise a question about how access should be managed after a generator leaves part of its permission unused. A smaller export allowance can be fully used yet permit less electricity to reach the network. Our findings show why operators should assess whether changing access lets other generators deliver more, and whether later restrictions preserve export when output recovers. Physical feasibility, delivered electricity and the distribution of access must therefore be considered together. With reliable estimates of available output, operators can assess which request reductions would enable more delivery. Regulators can then weigh the aggregate export gain against restrictions on individual generators within existing network limits.

INTRODUCTION

Homes and businesses with rooftop photovoltaics (PV) can meet part of their own electricity demand, but exporting the surplus requires access to a shared distribution network. Voltage and equipment limits constrain that access. Reinforcement and flexible power-flow control can expand the operating range[1]; allocation determines how generators share access within that range. China's distributed-PV management framework accordingly places connection capacity and operational control alongside project development[2]. Making effective use of existing export capacity therefore complements new generation and network reinforcement.

Flexible export arrangements make this allocation an ongoing operating task. South Australia's flexible-export programme offers an alternative to restrictive fixed limits[3], and California's limited generation profiles permit export schedules aligned with time-varying hosting capacity[4]. The European Union also provides for flexible connection agreements where network capacity is limited[5]. Each provides a way to vary permitted export with network conditions, although their procedures differ. Operating envelopes express this permission as connection-level bounds, allowing owners or aggregators to schedule their resources within network constraints[6–8]. Owners retain control over output within their bounds, so the allocation's use depends on the generation available at each connection. Permission can remain unused at one location while another generator has electricity that its limit prevents it from exporting.

Reallocation helps only when additional permission can reach a connection that has electricity to use it. Both electrical location and available output matter: the same increase in export can have different effects on voltage and equipment loading at different connections. Reducing unused permission helps total delivery only if the resulting allocation recovers more delivery than it removes. Simply making every allowance small enough to be fully used can sacrifice export. An operator considering future restrictions needs to identify the delivery that an alternative allocation could recover.

We study arrangements in which participants request access before their output is known. A related information sequence appears in Project Converge's shaped-operating-envelope design: aggregator bids and customer contribution plans precede the calculation of export bounds[9]. In our setting, an award is an export ceiling that remains fixed within the interval; unused access is not reassigned before the interval ends. A request above the point forecast can protect export when output is high, yet leave permission unused when output is low. If the larger award reduces another generator's useful access, the requester's gain can be smaller than the loss elsewhere. Assessing this trade-off requires available output at all affected connections.

Figure 1 illustrates the distinction. Connections A and B share a binding limit, and B leaves the same amount unused in (b) and (c). Reallocation increases delivery only in (c), where A has additional available output. B's identical award and delivery cannot distinguish these cases. A's delivery at its limit also fails to reveal how much more it could have supplied. The missing information is the output that could have been exported under an alternative feasible award.

Existing research addresses network feasibility, allocation objectives and participant incentives. Robust operating envelopes account for uncertain injections when defining network-feasible bounds[10,11], while network-secure bidding combines those bounds with adjustments to uncertain output[12]. Allocation objectives can favor total access, proportional sharing or equality[13–16]. Market mechanisms coordinate bids and prices with network constraints and participant incentives[17–19]. These approaches use information available before delivery to determine access. Project EDGE's comparison of forecast update frequencies shows why the timing and accuracy of that information matter[20]. Completed-interval records can answer an additional question:

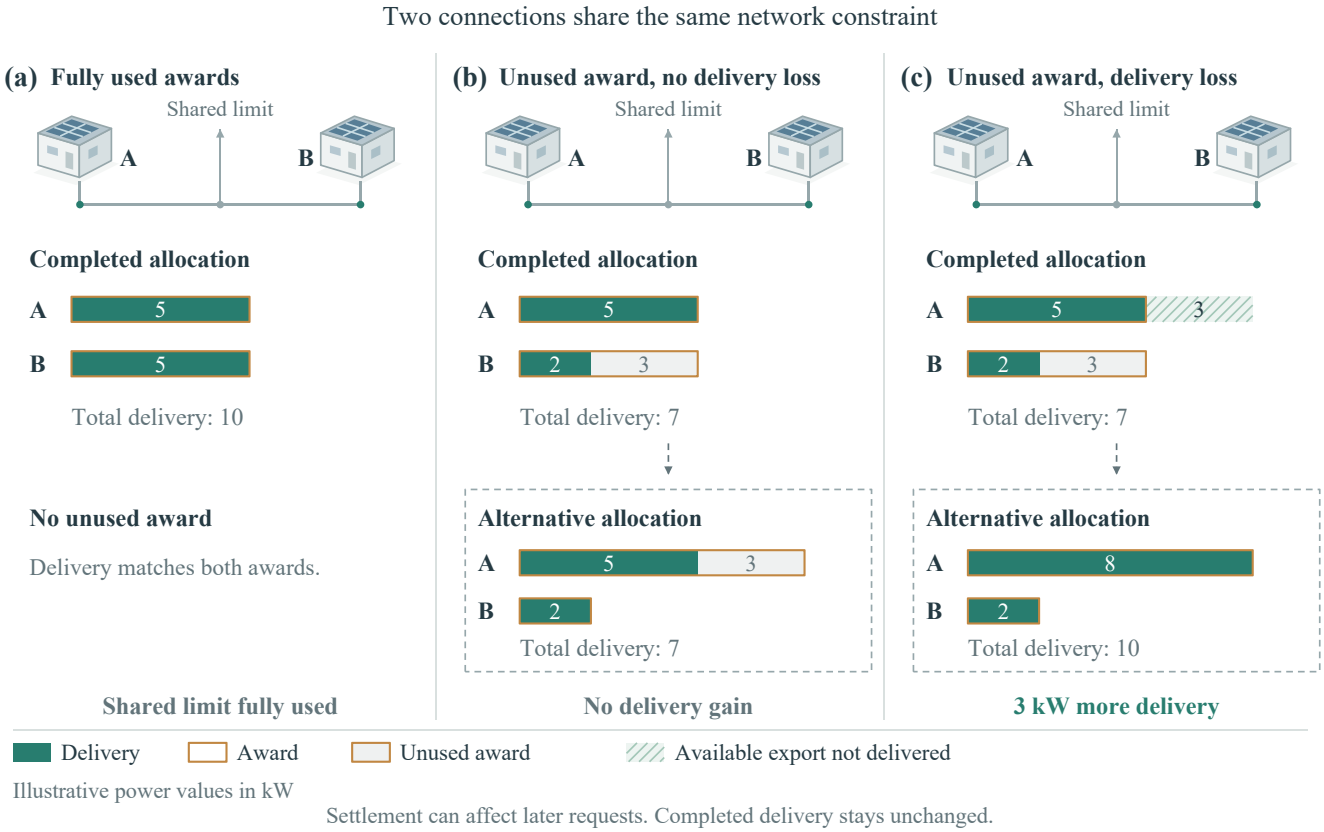


Figure 1. When unused export permission displaces delivery. Two connections share a binding network constraint. (a) Both awards are fully used. (b) B leaves part of its award unused, but A has no additional available export. An alternative allocation increases A's award without increasing delivery. (c) B leaves the same portion unused while A has available export that its award prevents it from delivering. The alternative allocation increases total delivery. Teal fills show delivery, amber outlines show awards, and hatching shows available export not delivered. Numbers within each segment give its power in kW. Dashed frames show calculations for the completed interval. Settlement can affect later requests but cannot restore that interval's lost delivery. The illustrative values use a shared 10 kW authorization bound and initial awards of 5 kW each; they are not case-study results.

whether a particular reallocation would have recovered delivery under the conditions that actually occurred.

History-dependent allocation and access exchange are particularly relevant to this question. Fair-over-time coordination spreads curtailment across participants and intervals[21]. Stateful pricing and allocation uses historical delivery relative to available export to adjust subsequent priorities[22]. SecuLEx instead permits network-checked exchange of issued limits according to customer needs[23]. These approaches establish ways to redistribute access within or across intervals. The question here is which unused awards provide a reason to restrict later requests. A history of under-delivery alone cannot answer it: identical shortfalls can leave another generator constrained or leave no useful export blocked. We test the alternative allocation under the completed interval's conditions. The test asks whether the gain in delivery elsewhere exceeds any loss from the group whose requests are reduced, while the network remains feasible. The proposed restriction thus depends on recoverable delivery, not on the shortfall alone.

Settlement changes the incentive to request access. A charge based only on B's shortfall gives the same signal in Figure 1(b) and (c), although only one contains recoverable delivery. A charge informed by system-wide delivery effects would also need to estimate the export lost elsewhere. Faster reassignment can address a mismatch before the interval ends. Where awards

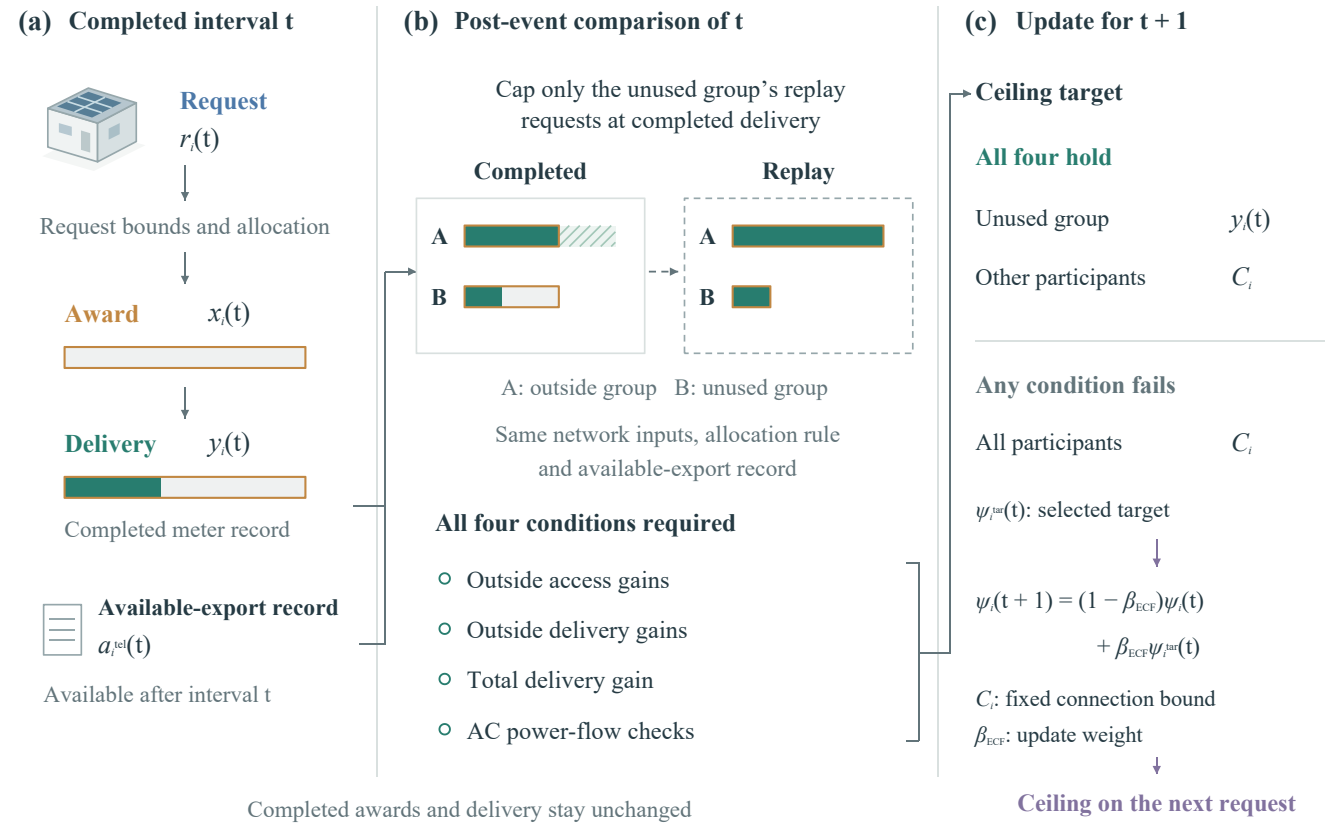


Figure 2. Using completed delivery to update later request ceilings. (a) Connection and feedback bounds limit requests before the existing rule allocates access. (b) The unused-award group's requests are reduced to completed delivery, and allocation is repeated under the same exogenous network conditions. Available-export estimates test delivery recovery. Qualification requires more outside access and delivery, higher total delivery and AC feasibility. (c) A qualifying comparison lowers the group's next request ceiling; all other ceilings recover toward connection bounds. If the test fails, all ceilings recover. Completed awards and delivery stay unchanged.

remain fixed, completed records instead offer a basis for improving later allocations, consistent with measurement-based approaches to power-system coordination[24,25]. Two questions follow: which completed shortfalls contain recoverable delivery, and whether acting on that information improves later delivery. The second does not follow from the first: output may recover after the interval that prompted a restriction.

We examine these two questions using one uncertain-output model for both request choice and subsequent delivery. A larger request can increase the requester's expected payoff and unused award while reducing total delivery, which we call a request–access–delivery externality. The same model also produces large requests that protect useful export. Whether a larger request displaces delivery depends on the awards and available output at all affected connections.

Externality-conditioned counterfactual feedback (ECF) tests completed allocations to update future request ceilings (Figure 2). After an interval ends, it reduces the unused-award group's requests to completed delivery and recalculates awards and power flows. The allocation rule, topology, load and network limits stay fixed. Estimating the resulting delivery requires available-export records that remain valid under the alternative awards; metered delivery alone cannot reveal output blocked by a limit. ECF lowers the group's next request ceilings only if the feasible reallocation raises delivery elsewhere and in total. Other ceilings recover toward connection bounds, and all recover if the test fails. Completed awards stay unchanged.

The experiments distinguish the value of this test from simply reducing requests or updating less often. Across five allocation rules, ECF preserves most of the larger-request group's delivery gain and exceeds proportional reductions of the same total request. In the Australian public-data simulation, it also outperforms random updates retrospectively calibrated to a similar frequency. Replacing ceilings outright with the latest targets, however, lowers delivery below no feedback in both the controlled zero-cost comparison and the public simulation. The distinguishing step is to test whether reallocating unused awards would increase total delivery before restricting later requests. The fixed feedback gain controls how strongly this result changes the next request ceiling.

RESULTS

Larger requests can protect one generator's export while causing a greater loss elsewhere. The controlled experiments show how targeted reductions recover part of that loss. Public-data simulations then distinguish the value of choosing update intervals from the benefit of updating less often. In both studies, stronger restriction can reverse the delivery gain. Table 1 identifies the alternative explanations tested by each control.

Table 1. How the comparison mechanisms differ. C denotes the controlled study and A the Australian simulation. Paired controls use the same network and available-export trajectory.

Mechanism	What changes	Question tested
No feedback (C, A)	Ceiling stays at the connection bound.	Baseline delivery.
Shortfall cost (C)	A modeled cost changes the initial request.	Effect of an incentive before allocation.
Mismatch-only feedback (C)	A participant's own delivery-to-award ratio updates its ceiling.	Effect of responding to shortfall alone.
Update without replay test (A)	ECF's formula applies after every unused award.	Effect of removing the test, including more frequent updates.
Frequency-calibrated random update (A)	The same formula updates randomly selected unused-award intervals.	Whether a similar update frequency explains ECF's gain.
Equal-total reduction (C, A)	ECF's total reduction is spread proportionally.	Value of where requests are reduced.
Previous-day persistence (A)	Request uses the previous-day output estimate alone.	Cost of a conservative request.
ECF (C, A)	Qualifying unused awards lower later ceilings.	Combined effect of testing and updating.

Private delivery gains can accompany a larger loss elsewhere

Larger requests increase the requester's expected delivery but reduce total delivery on average when there is no shortfall cost. Across 1,500 combinations of participant location, allocation rule, reactive-power setting and forecast uncertainty, the requester gains 5.2 kW relative to a point-forecast request. Other participants lose 18.8 kW. Total delivery falls by 13.5 kW, with all reported differences calculated before rounding. The same uncertain-output model generates both the request choice and delivery, linking this private benefit to the larger loss elsewhere.

Figure 3 traces the request choice as forecast uncertainty σ and normalized shortfall cost ϑ change. The cost parameter is the modeled charge for one unit of unused award divided by the value of one unit delivered. We test five allocation rules—weighted proportional (WP), equal reduction (ER), equal access (EA), maximum export (ME), and fairness-aware allocation (FA)—under two reactive-power settings. Each larger request is compared with the same participant's forecast-based request. The output model is specified in Experimental procedures.

Individual cases differ from this mean. We classify a request–access–delivery externality when the requester's expected payoff and unused award rise while total delivery falls. A productive large request raises payoff without a total-delivery loss beyond the numerical tolerance. This classification concerns the effect of choosing a larger request; ECF instead tests whether a completed unused award could have supported delivery elsewhere. The definitions appear in Experimental procedures and supplemental section S2.

At $\vartheta = 0$, 71.5% of the 1,500 cases meet the externality definition, while 11.9% are productive large requests. The other 16.6% either retain the forecast-based request or meet neither classification. Thus request size and unused access alone cannot identify the delivery outcome. These shares pool all five uncertainty levels; Figure 3(c) fixes $\sigma = 0.20$ to show how the categories vary with shortfall cost.

A higher shortfall cost discourages requests likely to leave unused awards, but one charge cannot represent every loss to others. Panel (d) divides the additional outside-delivery loss by the additional unused award for a small increase in the request multiplier. Among steps that raise own delivery and unused awards while lowering total delivery, the ratio $\vartheta^{\mathrm{ext}}_{i,\Delta}$ has a pooled median of 2.29; its rule-specific medians differ substantially. Identical unused amounts can therefore accompany different losses elsewhere. Supplemental section S2 gives the full cost comparisons and ratio distribution. We next test whether feedback can reduce the outside loss while preserving useful export by the requester.

Targeted request reductions preserve useful export access

At zero shortfall cost and the intermediate feedback gain, $\beta_{\mathrm{ECF}} = 0.5$, ECF delivers 24.8 kW more on average across the 16-interval trajectories than proportional reductions of the same total request. The following comparisons separate the effects of the initial request choice, the placement of reductions and the strength of the update. Feedback controls share raw requests and network conditions; a shortfall cost instead changes the request choice before allocation. Figure S1 follows one trajectory as output falls and recovers.

A shortfall cost reduces displacement but also useful delivery

A shortfall cost changes requests before anyone receives an award. Raising ϑ from 0 to 1 reduces the mean multiplier from 1.51 to 1.14 and the share of intervals meeting the economic externality definition from 50.1% to 15.6%. But it also removes 45.6% of the larger-request group's delivery gain (Figure S2). At $\vartheta = 10$, none of these intervals meets the externality definition, while the group's gain is almost zero. The charge reduces delivery displacement but also removes permission that the group would use at high output.

At $\vartheta = 0$, ECF improves total delivery by 24.9 kW over no feedback while retaining 79.4% of the larger-request group's original delivery gain. This recovery changes the comparison with forecast-based requests from a 21.7-kW loss without feedback to a 3.2-kW gain with ECF. Mismatch-only feedback, which updates each ceiling from that participant's delivery-to-award ratio, recovers 15.2 kW relative to no feedback. ECF therefore delivers 9.7 kW more than this

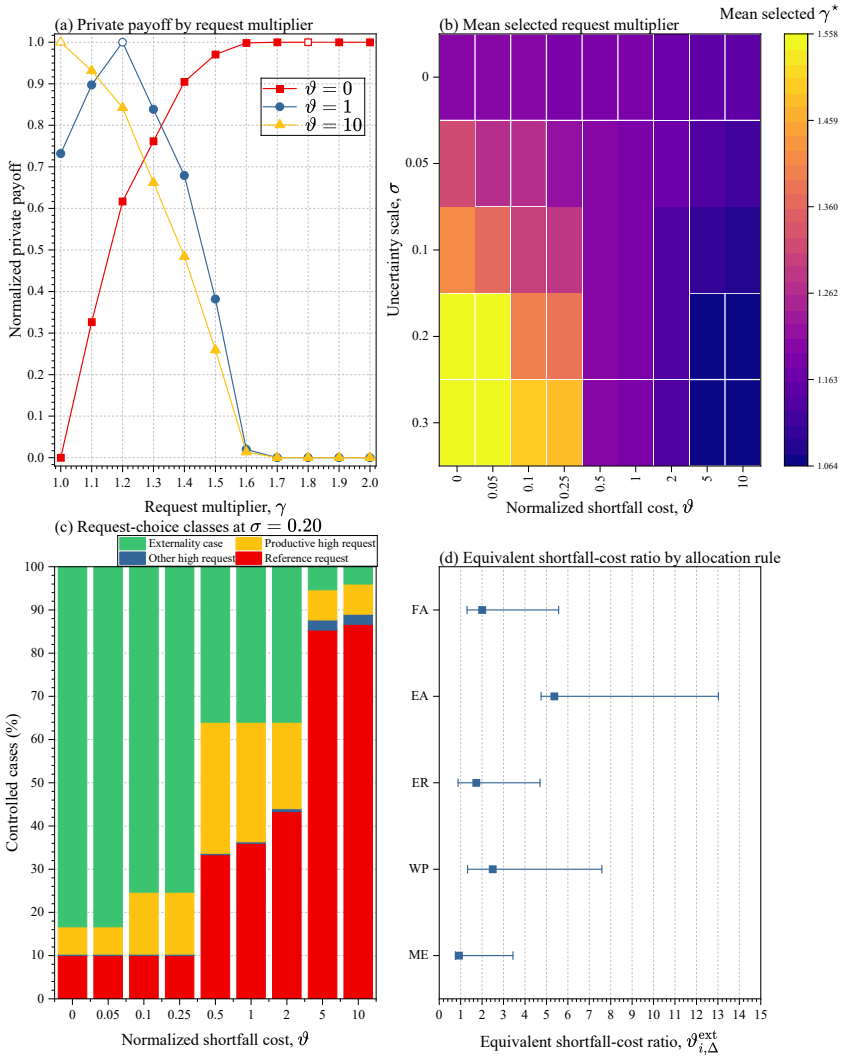


Figure 3. A larger request can protect own delivery or reduce total delivery. (a) Private-payoff curves for participant 21 under WP, Q0 (zero participant reactive power) and $\sigma = 0.20$, at three normalized shortfall-cost levels. Within each curve, payoff is rescaled as $(U - U_{\min})/(U_{\max} - U_{\min})$; a zero span is plotted as zero. (b) Mean selected request multiplier for each uncertainty and shortfall-cost setting. Means cover participants, rules and reactive-power settings. (c) Shares of four request classes at $\sigma = 0.20$ ($n = 300$). Reference requests use the point forecast. Productive high requests raise payoff without a total-delivery loss beyond the classification tolerance. Request–access–delivery externality cases raise payoff and unused awards while lowering total delivery. Other high requests meet neither definition. (d) Finite-difference cost ratio $\vartheta^{\text{ext}}_{i,\Delta}$: expected delivery loss elsewhere per unit increase in expected unused award for a 0.1 request-multiplier step. Squares show rule-specific medians, and bars show interquartile ranges. WP, ER, EA, ME, and FA denote weighted proportional, equal reduction, equal access, maximum export, and fairness-aware allocation, respectively.

shortfall-based update. Supplemental section S3 reports the group decomposition and other cost settings. At $\vartheta = 10$, ECF raises mean delivery but increases the economic externality share from zero to 1.3%. A higher mean delivery can therefore coexist with more intervals classified as request–access–delivery externalities.

Where requests are reduced determines recovered delivery

To isolate the effect of where requests are reduced, the equal-total control removes the same total request as ECF in each interval but distributes the reduction proportionally across con-

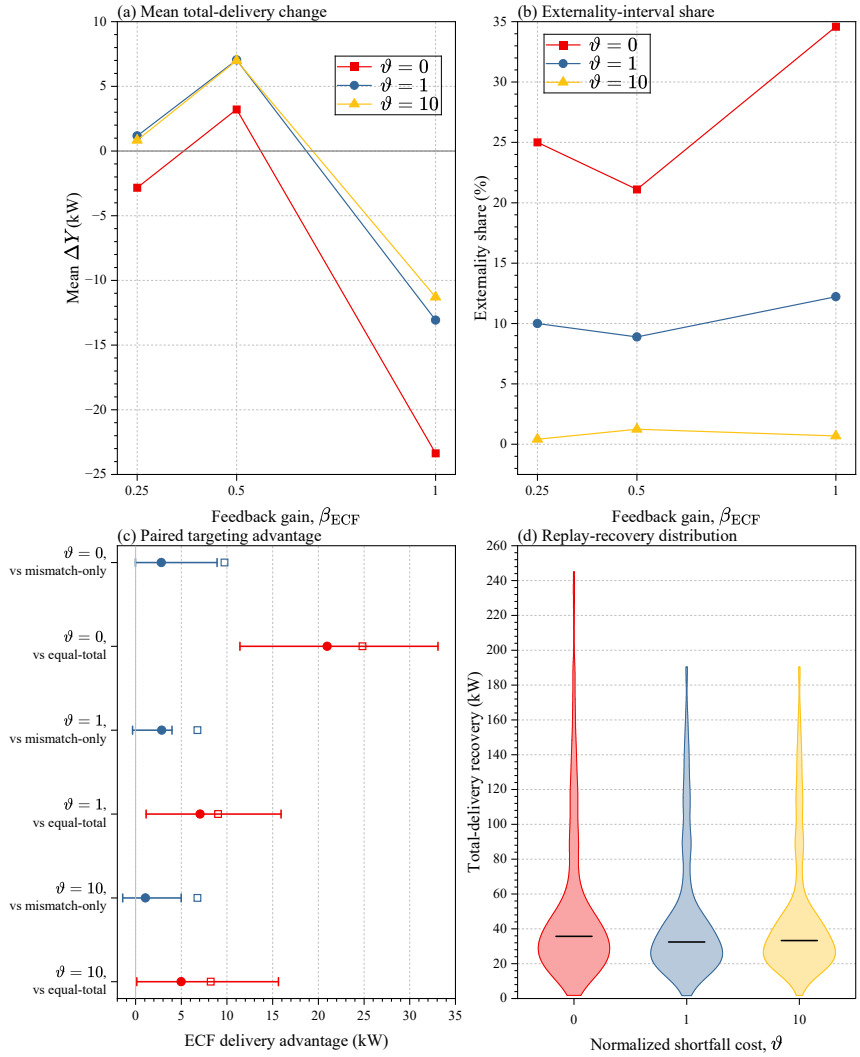


Figure 4. Targeted feedback preserves delivery, but a unit-gain update reverses the improvement in the controlled cases. (a) Mean signed change in total delivery relative to forecast-based reference requests and (b) externality-interval share for the three tested feedback gains. The series denote $\vartheta = 0, 1, 10$. (c) ECF delivery advantage in matched trajectories. Rows labeled "vs mismatch-only" compare ECF with mismatch-only feedback. Rows labeled "vs equal-total" compare ECF with the equal-total control. Panel (c) uses the primary gain, $\beta_{\mathrm{ECF}} = 0.5$. Filled circles show medians, open squares show means, and whiskers show interquartile ranges. The gray line marks zero. (d) Distribution of total delivery recovered when the replay meets all four conditions. Violin width shows empirical density, and the black horizontal segment shows the median.

nections. ECF exceeds this control in 95.6% of the 90 matched trajectories at zero shortfall cost. Thus the mean advantage comes from where requests are reduced, rather than their total reduction. Supplemental section S3.5 reports comparisons at the other cost settings.

Selecting a useful interval is insufficient if the next ceiling is too tight. At zero shortfall cost, replacing the ceiling with one interval's target changes the 3.2-kW gain over forecast-based requests into a 23.4-kW loss (Figure 4). This unit-gain response can restrict export when output subsequently recovers. The intermediate gain of 0.5 gives the largest mean delivery among the three tested gains; the reversal also occurs at the other cost settings.

Almost all controlled replays find recoverable delivery (supplemental section S3.5). These trajectories reveal the importance of placement and update strength. The public-data comparison below tests interval selection when recovery opportunities are less frequent.

Output timing and allocation rules alter the delivery gain

The delivery advantage persists across the five allocation rules, but its size depends on how each rule distributes access. At $\vartheta = 0$ and $\beta_{\mathrm{ECF}} = 0.5$, ECF exceeds equal-total reduction by 9.7–50.8 kW on average across the rules (Figure 5). Mean delivery also exceeds no feedback under all five rules.

The three temporal profiles preserve each participant's available-export values and frequencies. They change their order and, because participants have different phase offsets, which output states occur together across connections. Clustered low output gives the largest mean gain, intermittent low output the smallest, and the shock–recovery sequence lies between them. The timing and coincidence of output therefore affect how much later delivery can be recovered, even with the same individual output frequencies.

The advantage also remains positive for one changed requester, five nearby requesters and five dispersed requesters. The five-participant sets differ in identity as well as location; their contrast reflects both. The two reactive-power settings give nearly identical mean gains. Supplemental section S3.5 collects the factor-level values and the allocation-equality results alongside Figure S3.

AC feasibility and allocation equality

The controlled award and delivery states satisfy nonlinear AC convergence, power-balance, voltage and two-ended apparent-power criteria within the declared synthetic operating envelope. Figure S3 and Section S3.5 report the loading and voltage margins.

More delivery need not mean more equal access. The capacity-normalized Jain index in Figure S3 measures equality of awards, not social fairness. At $\vartheta = 0$, the mean index is higher with ECF than with no feedback and equal-total reduction under all five rules. Four rules nevertheless remain below the forecast-based reference.

Public-data simulations show when targeting raises export

Household generation and demand histories provide a different test of feedback: unused permission and restricted generation need not occur together. We examine the energy gained over time, then test whether ECF's choice of update intervals adds value beyond reducing update frequency.

Better use of access increases delivery, not just utilization

Across 80 Australian network-days, ECF delivers 239.36 MWh of modeled export: 1.39 MWh (0.59%) more than either no feedback or equal-total request reduction (Figure 6). The simulation combines a published medium- and low-voltage network with public household PV and load histories, from which requests and available export are constructed. The primary comparison uses proportional allocation; Experimental procedures and supplemental section S10 describe the scenario construction.

The gain combines more authorization with less unused permission. Relative to no feedback, ECF authorizes an additional 1,206 kWh and leaves 186 kWh less unused. Together these changes account for the additional 1,392 kWh delivered, according to the identity in (8). Equal-total reduction has almost no delivery effect because proportional scaling preserves request

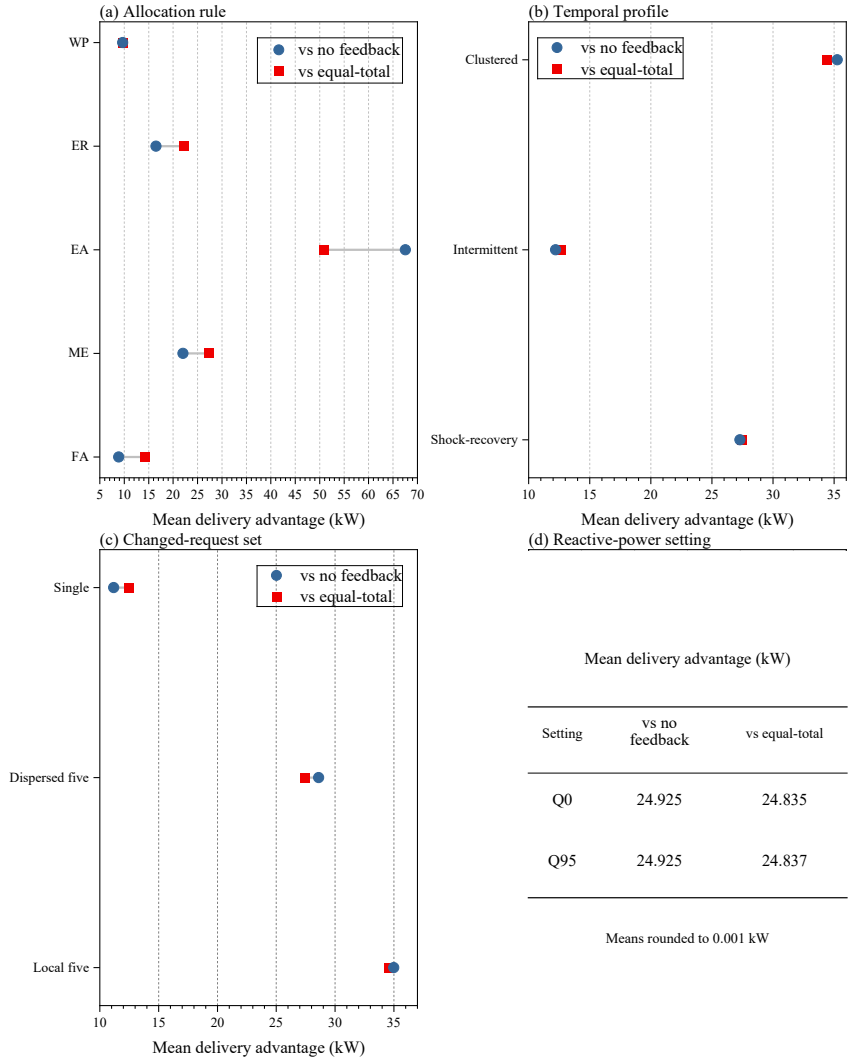


Figure 5. The controlled delivery advantage persists across allocation rules and output timing at $\vartheta = 0$ and $\beta_{\mathrm{ECF}} = 0.5$. Markers are factor-level means, grouped by (a) allocation rule, (b) temporal profile and (c) changed-request set. Panel (d) tabulates means for the two reactive-power settings to show their close agreement. Blue circles compare ECF with no feedback. Red squares compare ECF with the equal-total control. WP, ER, EA, ME, and FA denote weighted proportional, equal reduction, equal access, maximum export, and fairness-aware allocation, respectively. Q0 sets participant reactive power to zero, whereas Q95 uses 0.95-power-factor absorption. The temporal profiles preserve individual output frequencies but change ordering and cross-connection coincidence.

shares: a binding network limit can be met by changing the common allocation factor instead. Targeted feedback changes those shares and therefore the awards at the binding network limit. Supplemental sections S4 and S11.1 give the derivation and full energy balance.

The gain varies across days: 44 improve, 32 lose delivery and four remain unchanged relative to either control. Against no feedback, the mean is 17.4 kWh per day, with a dependence-adjusted 95% interval of 5.1–29.7 kWh. A continuous 35-day block contributes 1.06 MWh of the 1.39-MWh gain; the remaining dates give a positive total, but the interval for their mean daily difference includes zero (supplemental section S10.3). ECF selects a delivery-target reduction in 12.6% of half-hours, although previously tightened ceilings can remain below capacity between selections.

The additional connection-point delivery reduces net upstream import by 1.38 MWh relative to either control, about 98.9% of the connection-point gain. The feeder remains a net importer; sup-

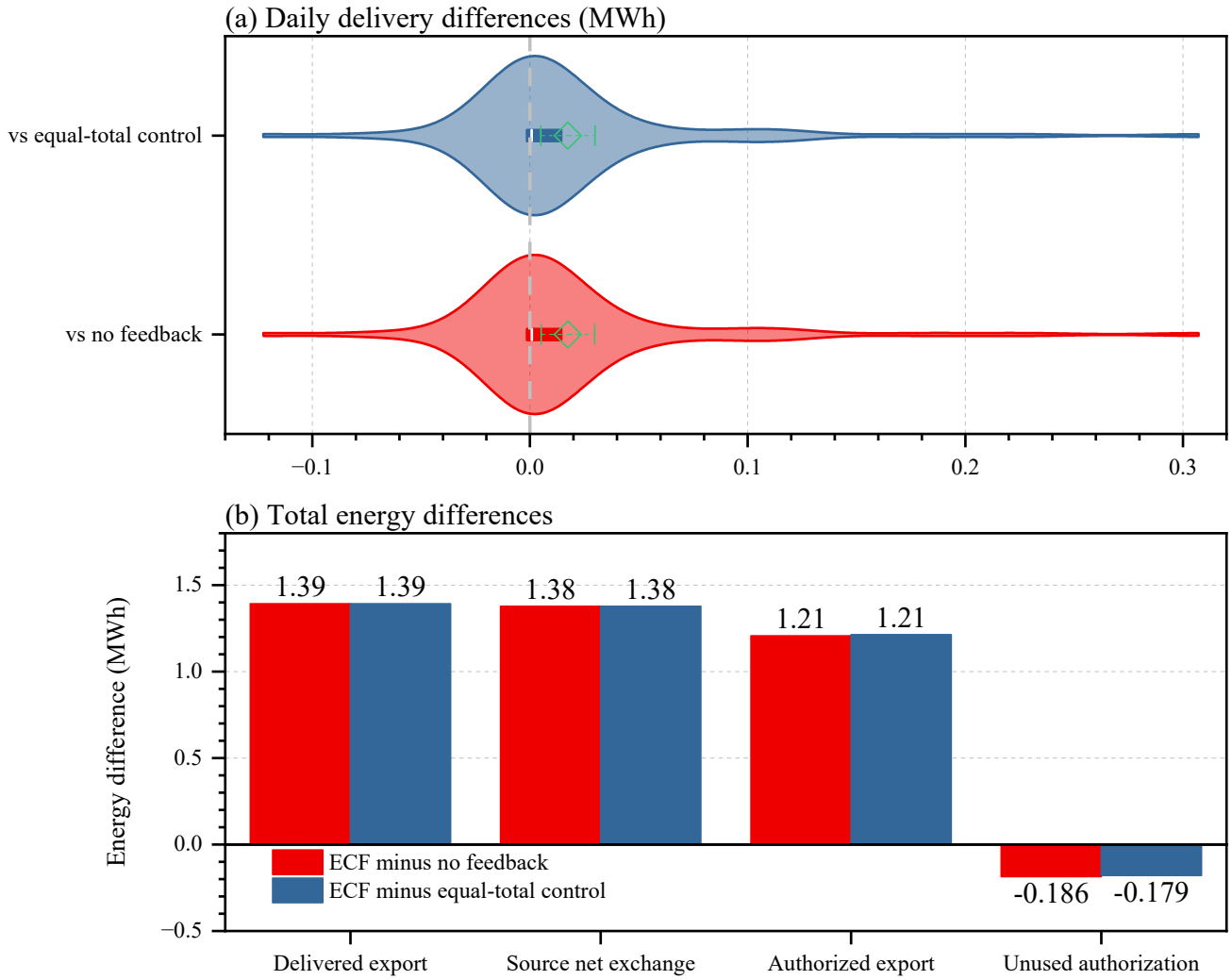


Figure 6. Modeled delivery gains and their energy balance in the Australian public-data simulation. Available export is constructed from PV and load profiles. (a) Paired daily delivery differences for ECF against no feedback and the equal-total control. Violin width shows empirical density. Dark boxes span the 25th–75th percentiles, and white ticks show medians. Green diamonds show means; capped green lines show 95% intervals for the mean daily difference, adjusted for short-range dependence between neighboring dates. (b) Aggregate 80-day differences in delivered export, source-terminal net exchange, authorization, and unused authorization. Each difference is ECF minus the indicated control. Delivery equals authorization minus its unused portion. Here the delivery gain accompanies increased authorization and reduced unused authorization. Positive source-terminal differences indicate reduced net upstream import over the assessed period.

plemental section S11.1 gives the circuit loss and passive-element terms. By contrast, previous-day persistence uses 89.0% of its authorization but delivers 50.33 MWh less than ECF. It omits the forecast increment and therefore requests too little on many high-output intervals. Utilization alone rewards a small, fully used award even when more electricity could have been delivered.

The public-study trajectories satisfy the stated allocation, customer-voltage and transformer-loading criteria (supplemental section S14.3).

Selecting recovery opportunities outperforms replay-free updates

ECF both selects update intervals and updates less often than a response to every unused award. Two controls examine the value of selection beyond the lower update frequency. We first remove the replay test but keep the same ceiling formula and gain. Updating after every unused award lowers modeled delivery by 6.42 MWh relative to ECF, with ECF ahead on 75 of 80 days. This control removes more unused permission but also removes useful export (Table 2).

That comparison also makes updates much more frequent. The second control keeps the target and gain but randomly selects unused-award intervals. Twelve schedules are retrospectively calibrated to approximately ECF's update frequency. ECF exceeds every schedule and delivers 1.34 MWh more than their mean, despite leaving almost the same total authorization unused. The mean paired daily advantage is 16.8 kWh, with a dependence-adjusted 95% interval of 7.8–25.8 kWh. A similar update frequency is therefore insufficient to reproduce ECF's gain. Replay-informed selection preserves more useful access than these random schedules; their separate delivery histories also produce different restriction magnitudes and durations. Supplemental sections S11.2–S11.3 give the control design and full results.

Table 2 collects the corresponding 80-day energy totals for each update rule.

Table 2. Modeled export over the same 80 Australian network days. The update without a replay test uses ECF's ceiling formula and gain after every unused award. The frequency-calibrated random row averages 12 replay-free schedules with the same formula and gain. The equal-total control matches ECF's request reduction in each half-hour. All energy totals are in MWh and are rounded independently.

Control	Authorized	Unused	Delivered
No feedback	328.490	90.526	237.963
Equal-total reduction	328.483	90.519	237.964
Update without replay test	305.159	72.226	232.933
Frequency-calibrated random update	328.357	90.345	238.012
ECF	329.696	90.341	239.356

Figure 7 shows how the advantage over similarly frequent random updates is distributed across days and control trajectories.

Update strength and record error can limit the benefit

Selecting an interval is only part of feedback: the replay concerns completed output, whereas its ceiling applies to later output. Unit gain loses 3.04 MWh relative to no feedback, reversing the moderate-gain advantage (Figure 8). Replacing the ceiling with one interval's target can deny export after output recovers. Retaining ceilings across consecutive days preserves both the moderate-gain advantage and the unit-gain loss (supplemental sections S12.1–S12.2).

Scaling the available-export record, delaying it or replacing missing values with completed delivery can change individual replay decisions. Each tested perturbation nevertheless leaves a positive mean delivery advantage over no feedback and equal-total reduction. These tests change the recovery calculation while AC checks retain unperturbed profile-derived availability (supplemental section S12.1).

A separate test also uses the perturbed record for the physical decision screen, with an independent check against unperturbed availability. Using a delivery lower bound avoids updates unsupported by reference recovery, but underestimation makes the test more restrictive. At the

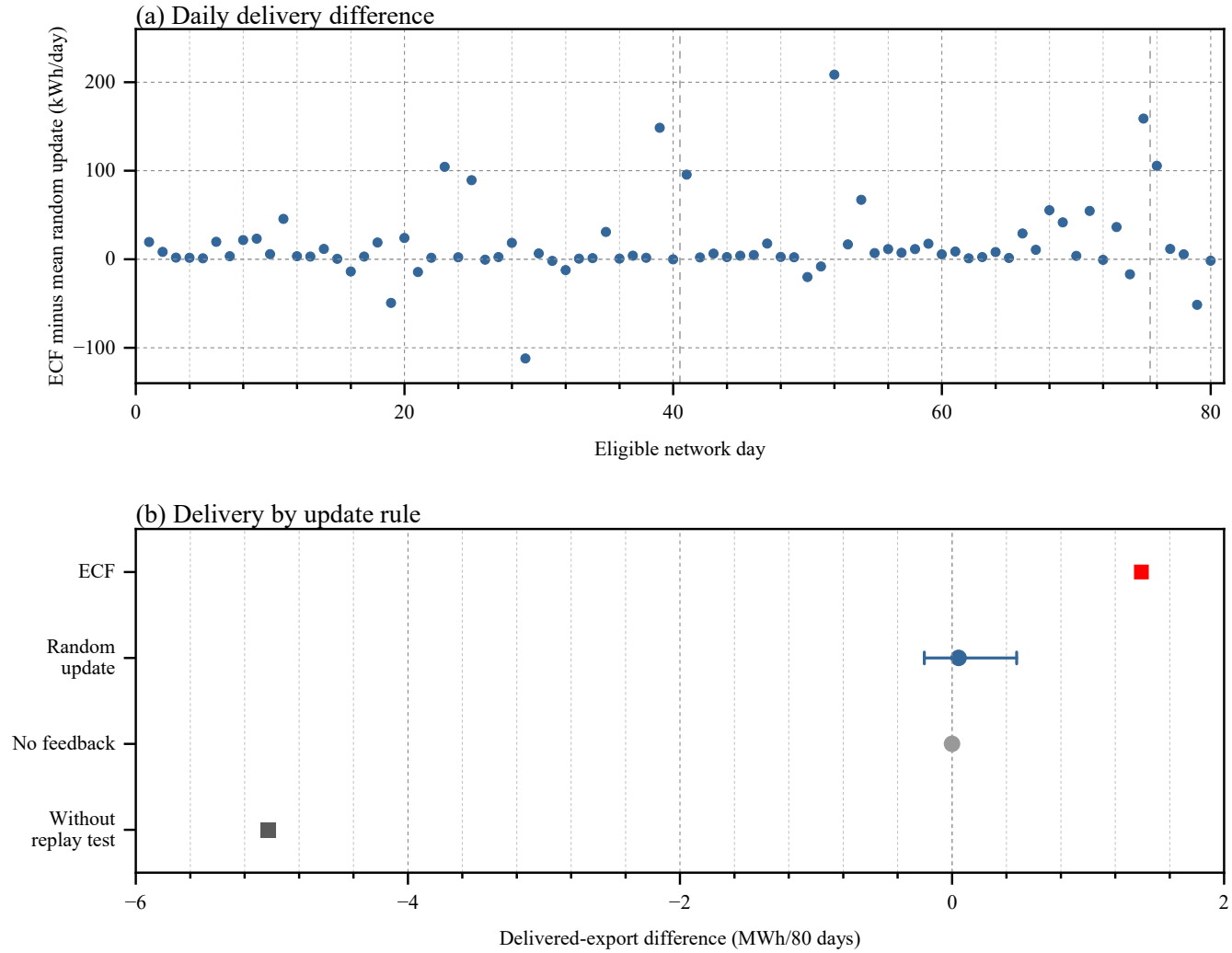


Figure 7. Replay-selected updates recover more modeled delivery than comparably frequent random updates in the Australian simulation. (a) Paired daily modeled-delivery difference between ECF and the mean of 12 frequency-calibrated random updates. The vertical dashed lines bound the longest continuous 35-day block. (b) Eighty-day delivery relative to no feedback. The blue marker is the mean across 12 fixed random schedules; its horizontal range spans their individual totals. The red square marks ECF; gray markers denote no feedback and updating after every unused award. The number of half-hours selected for a delivery-target update is 483 for ECF, 463–519 for the random schedules, 0 for no feedback and 3,086 for updating after every unused award. Ceiling recovery still operates in other intervals. The schedules are retrospectively calibrated to approximately ECF's update frequency, without using replay to choose their intervals. Their independent ceiling histories need not give identical update magnitudes or durations.

largest negative bias, it delivers 269 kWh less than the point-record controller by missing useful recoveries; both still exceed no feedback. At the prescribed positive bias, subtracting the error radius recovers reference availability by construction. Supplemental section S12.3 gives the error model and contrasts. Within the assumed error bound, this test prevents overestimated availability from being mistaken for recoverable delivery. It can, however, exclude useful updates when output is already underestimated.

Spatial placement and available generation shape the benefit

Changing where the same household profiles are placed changes the gain from about 0.9 to 5.1 MWh over no feedback and equal-total reduction across five assignments. All five pointwise

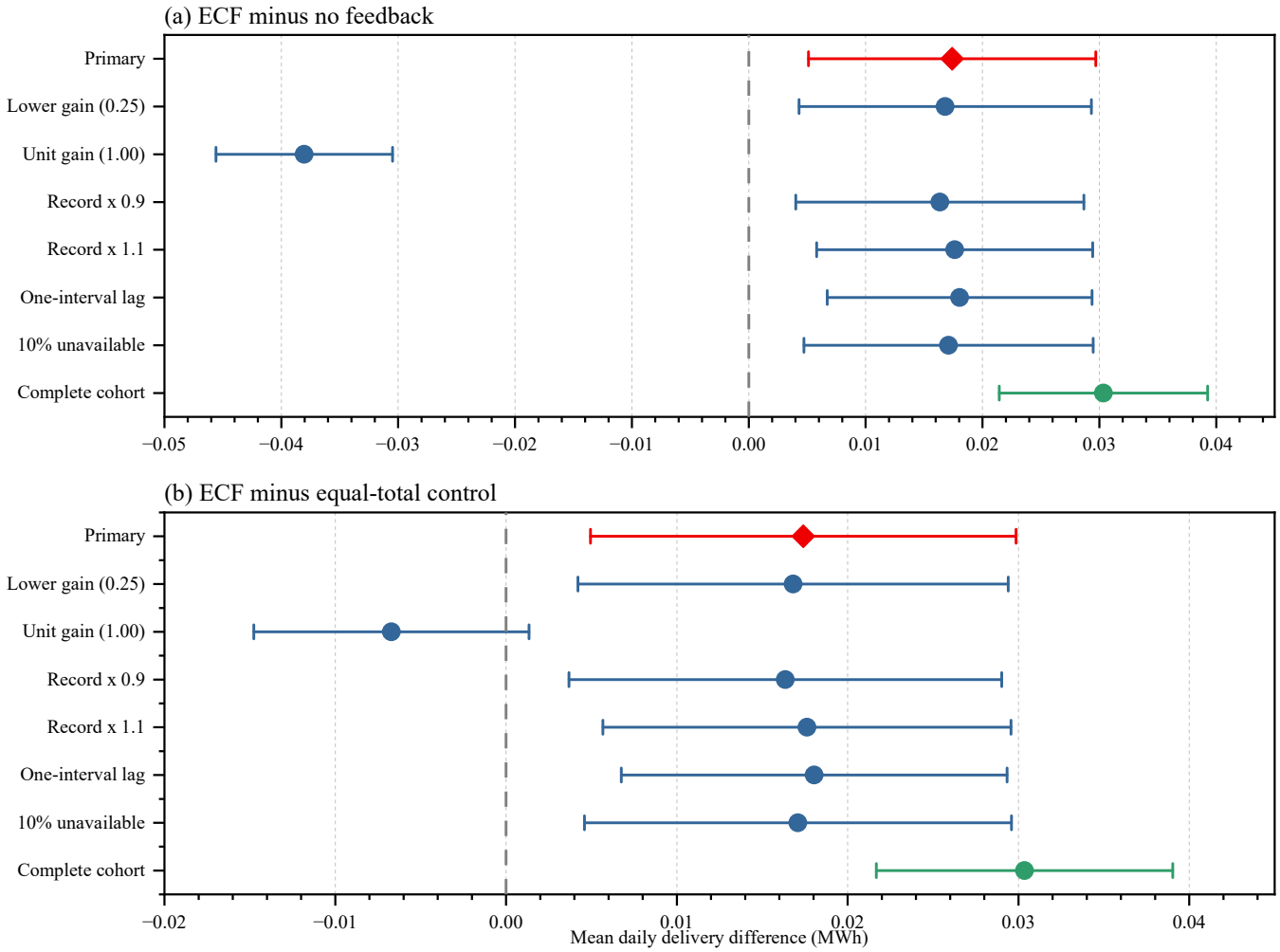


Figure 8. Gain, record quality and customer coverage change the Australian delivery result. Markers show paired mean delivery differences per network day. Bars show 95% intervals adjusted for short-range dependence between neighboring dates. (a) ECF minus no feedback. (b) ECF minus the equal-total control. The red diamond denotes the primary unperturbed-proxy result, and blue circles denote gain or record changes. Green markers denote the complete-customer cohort, which uses the same load multiplier but changes customer composition and calendar coverage (361 days rather than 80). Every transformed available-export record is floored at modeled delivery. The zero line separates positive from negative mean delivery differences.

mean-difference intervals remain positive. Varying PV capacity and connection bounds at fixed locations also preserves a positive gain, but the gain does not increase monotonically with installed PV. Thus more generation alone does not determine the benefit; it matters where access can be used. Supplemental sections S13.1–S13.2 give the matched settings, energy balances and uncertainty intervals.

Changing the network itself reveals both gains and null outcomes. In 13 LV sections of the parent feeder, ECF delivers more than equal-total reduction in every section and more than no feedback in 12; LV26 loses 3.7 kWh against no feedback. Of nine representative published LV models, six gain delivery and three have no qualifying interval and remain unchanged (Figure S5). The positive differences include a near-zero result. Supplemental sections S13.3–S13.4 report how useful reallocation varies across these networks under the common scenario construction.

The complete-customer cohort extends the comparison to 361 days at the same load scale. ECF gains 30.4 kWh per day over no feedback and equal-total reduction, with positive mean-difference intervals. This extension changes customer composition together with calendar coverage (supplemental sections S10.3 and S12.1).

Individual outcomes and request adjustment

The aggregate gain includes both individual gains and losses. Among 418 participants in the primary feeder, 330 deliver more than without feedback over the 80 days and 88 deliver less; the largest individual loss is 7.3 kWh (supplemental section S14.1).

The preceding public-data comparisons use a fixed forecasting rule to construct requests. We also test participants who adjust request-choice probabilities from their own delivery histories. ECF delivers 1.44 MWh more than no feedback and 4.53 MWh more than the same ceiling update without a replay test. Each control develops its own request history through experience-based request adjustment, specified in supplemental section S14.2.

DISCUSSION

The central finding is that less unused permission does not necessarily mean better use of a distribution network. A ceiling update can remove idle awards yet restrict generation that would otherwise have been delivered. ECF instead asks whether the unused award prevented export elsewhere, using an alternative allocation for the completed interval. Applying this test before updating requests increases delivery in the primary comparisons without changing the network limits or underlying allocation rule.

The controls explain why this test matters. Spreading the same total request reduction proportionally delivers less than targeting the selected connections. Random updates at a similar frequency also fail to reproduce ECF's public-simulation gain, despite leaving almost the same total authorization unused. The advantage thus depends on which requests are reduced and when, rather than solely on the amount removed or frequency of updating. Different ceiling histories also change restriction magnitude and duration in the random comparison.

The primary Australian gain over no feedback is 0.59%, and almost all of the additional export reduces upstream import. The change therefore extends beyond the allocation calculation to energy exchanged at the feeder boundary. This aggregate gain accompanies larger differences between feedback designs: updating after every unused award removes more idle permission but loses useful export. Reducing unused permission is therefore an inadequate performance target unless the update also preserves access that generators can use later.

These comparisons suggest specific criteria for evaluating flexible-access trials. Operators should compare net delivered electricity, not only the amount of permission left unused. This comparison must subtract any delivery lost by the restricted group from the gain elsewhere and follow delivery as output recovers. Regulators would also need to examine which participants lose export and how long their requests remain restricted. An aggregate gain alone cannot establish that an access rule distributes restrictions acceptably. These criteria complement forecasting and settlement where operators have authority to revise request ceilings. Where access can instead be reassigned within an interval, that faster response should be included in the comparison.

The response must also leave room for changing output. Replacing a ceiling with one interval's low delivery can prevent export when output recovers; unit gain produces this reversal in both the controlled zero-cost comparison and the Australian simulation. Update gain therefore needs to be assessed against changing output, rather than selected from replay recovery alone. Record uncertainty introduces a related trade-off: a validated lower bound protects the recovery calculation from overestimation, but can reject useful updates when the record already underestimates output.

ECF qualifies a joint reduction for the unused-award group. It does not identify each member's contribution to recovery, so the result alone cannot justify an individual penalty. The observed participant losses also matter for the acceptability of a group update, even when aggregate export rises. ECF's mean advantage persists when participants adjust requests from their own delivery histories. This result describes experience-based adjustment rather than strategic anticipation of future ceilings.

The two studies provide different physical tests. The controlled 141-bus case checks nonlinear AC voltage, balance and two-ended apparent power under synthetic branch ratings. The Australian models combine measured household histories with published network structures; requests and available export are constructed. Their physical checks cover customer voltage and native transformer ratings, with line ampacities unavailable. Applying the replay requires linked requests, awards, metered delivery and verified available export for the same connections and intervals. If an alternative award changes availability through storage, inverter control or flexible demand, the replay must include that response.

Safe export limits define what the network can accommodate; they do not by themselves ensure that generators can use the allocated permission. Completed-interval comparisons reveal where an alternative allocation could have supported more delivery. In the simulations, using this information to adjust later requests increases export when the update leaves room for output recovery. The practical implication is to evaluate access updates by net electricity delivered over time and by the restrictions imposed on individual connections, rather than by unused permission alone.

EXPERIMENTAL PROCEDURES

Resource availability

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Zhenning Pan (scutpanzn@163.com).

Materials availability

This study did not generate new physical materials.

Data and code availability

- Figure source data are included in Data S1. The project repository, `https://github.com/MarxHuang/ecf-export-access`, contains calculated results, input-source identifiers, preprocessing instructions, date selections and network-profile mapping tables. The source 141-bus network, CSIRO/GridQube network collection (doi:10.25919/ghnz-bk28) and Ausgrid Solar Home Electricity Data remain with their original distributors under their respective terms; raw third-party records are not redistributed.
- Simulation, preprocessing and analysis code, parameter files, environment specifications and reproduction instructions are maintained in the same project repository. Its checks of saved results are distinguished from full simulation reruns, which require the external source datasets.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

Requests, awards and available export

The model separates three stages. Requests determine export awards before output is known. Available export then determines how much of each award is used. ECF acts after delivery, changing the ceiling applied to a later request.

Let $\mathcal{N}$ be the participating export connections and $\mathcal{G} = (\mathcal{B}, \mathcal{L})$ the distribution network, with buses $\mathcal{B}$ and branches $\mathcal{L}$. In interval t, participant i requests $r_i(t)$ before its available export $a_i(t)$ is known. Its fixed connection bound is C_i. Feedback changes a request ceiling $\psi_i(t) \in [0, C_i]$, initially C_i, so the admitted request is

$$r_i^{\mathrm{adm}}(t) = \min\{r_i(t), C_i, \psi_i(t)\}. \tag{1}$$

Allocation rule $\mathcal{R}$ then grants access $\mathbf{x}(t) = \Pi_{\mathcal{R}}(\mathbf{r}^{\mathrm{adm}}(t); \mathcal{F}_{\mathcal{M}}(\mathcal{G}, \mathbf{z}(t)), \boldsymbol{\eta}_{\mathcal{R}})$. Here $\mathcal{M}$ denotes the network model, and $\mathcal{F}_{\mathcal{M}}$ its equations and limits under exogenous data $\mathbf{z}(t)$; $\boldsymbol{\eta}_{\mathcal{R}}$ contains rule parameters, held fixed in the comparisons. Completed delivery is $y_i(t) = \min\{x_i(t), a_i(t)\}$, with $0 \le y_i(t) \le x_i(t) \le r_i^{\mathrm{adm}}(t) \le \psi_i(t) \le C_i$. Thus $x_i - y_i$ is the unused award. Awards remain fixed within an interval; access is not redistributed during that interval. Feedback modifies only the admitted-request input in a later interval, not the rule or network limits. These variables are interval-average powers. Multiplication by interval duration and summation give the reported energy totals. Table S1 collects the quantities.

The coordinator receives requests before allocation and observes delivery afterwards. It also requires a post-event available-export record $a_i^{\mathrm{tel}}(t)$ to estimate delivery under alternative awards. Participants' private output signals and conditional distributions are not inputs to the coordinator. The following assumptions specify the record and the conditions under which it can support that comparison.

Assumption 1 (Post-event available-export information)**.** *The comparison uses an available-export record a_i^{tel} for the completed interval. Its origin and time alignment must support the*

delivery calculation under an alternative award. In the controlled study the record is exact simulated availability; in the Australian study the primary record is the profile-derived proxy a_i^{prof}. Record-error tests vary the replay input a_i^{tel}, while unperturbed a_i^{prof} continues to generate modeled delivery. Neither record changes an award already issued.

Assumption 2 (Same-state replay)**.** *The intervention changes only admitted requests in the completed-interval comparison. Topology, exogenous network data, participants, rule parameters and underlying available export remain fixed. Awards and power flows are recalculated. If a different award would change availability through inverter controls, storage, demand or portfolio dispatch, those responses must instead enter the replay model.*

Request choice and delivery displacement

The controlled study links a participant's request to later delivery through one conditional distribution of available export. Participant i observes a private signal s_i, forms a point forecast a_i^{fcst}, and assigns conditional cumulative distribution function (CDF) $F_i(\cdot \mid s_i)$ to its available export. Let $r_i^{\mathrm{ref}} = a_i^{\mathrm{fcst}}$ be its point-forecast request and $r_i^{\mathrm{chg}} = \gamma_i r_i^{\mathrm{ref}}$, $\gamma_i \geq 1$, a larger request. Holding other requests and exogenous network data fixed, the participant values own delivery and faces a modeled shortfall cost $\vartheta \geq 0$ per unit of unused award, normalized by the value of delivered export:

$$U_i(\gamma_i; \vartheta) = \mathbb{E}\big[\min\{x_i(\gamma_i), a_i\} - \vartheta[x_i(\gamma_i) - a_i]^+ \mid s_i\big]\,. \tag{2}$$

Here $[v]^+ = \max\{v, 0\}$ is the positive part. When several multipliers give the largest expected payoff, we select the smallest on the prespecified grid Γ. The same conditional available-export distribution determines this choice and the subsequent realization, linking the incentive to request more access with the electricity subsequently available to use it. A static changed-request case is classified as a request–access–delivery externality if the changed group's expected payoff and unused award both rise while aggregate expected delivery falls beyond the numerical tolerance. A productive high request raises the group's payoff without such a delivery loss. Dynamic cases use realized counterparts. This economic classification is distinct from ECF's operational replay test.

Where the allocation map is differentiable and the available-export CDFs are continuous at the current awards, write $J_{ji} = \partial x_j / \partial r_i$ and $\bar{Y} = \sum_j \mathbb{E}[\min\{x_j, a_j\} \mid s_j]$ for expected total delivery. Here $F_j(x_j)$ abbreviates the conditional cumulative distribution $F_j(x_j \mid s_j)$. Then

$$\begin{aligned} \frac{\partial U_i}{\partial r_i} &= J_{ii}[1 - (1+\vartheta)F_i(x_i)], \\ \frac{\partial \bar{Y}}{\partial r_i} &= \sum_j J_{ji}[1 - F_j(x_j)]. \end{aligned} \tag{3}$$

The derivatives can have different signs because private payoff excludes delivery lost elsewhere. The numerical study uses a discrete output tree and can change active constraints, so all reported classifications use matched finite differences. When additional outside-delivery loss and additional unused award are both positive, their ratio measures the loss elsewhere per additional unit left unused. This is the quantity plotted in Figure 3(d). Its finite-change definition and relation to the local marginal shortfall cost are in supplemental section S2; that section also defines the capacity-normalized Jain index.

Allocation rules and physical checks

Five rules use the same admitted requests and controlled-study network domain (Table S2). Weighted proportional (WP) scales all requests by one feasible factor. Equal reduction (ER) subtracts a common amount; equal access (EA) applies a common cap. Maximum export (ME) maximizes total authorized access. Fairness-aware allocation (FA) trades authorized access against a request-normalized equality term. Their one-parameter formulas, conic objectives, deterministic tie-breaks and sensitivity derivations are specified in supplemental sections S5–S8. Section S4 derives the matched request-reduction comparison. All rules use an affine network approximation with second-order-cone constraints on two-ended branch loading. Nonlinear AC power flow separately verifies the resulting states.

The controlled medium-voltage case uses the published radial 141-bus network[26], a 0.70 demand scale and 30 fixed connection points. Fixed synthetic branch ratings define this mechanism experiment; they are not native equipment nameplates. Q0 denotes zero participant reactive power and Q95 denotes 0.95 power factor with reactive-power absorption. Each matched reference, changed-request, award, simulated-delivery and replay-delivery state is checked by nonlinear radial AC power flow at its fixed reactive-power setting. The check requires convergence, active- and reactive-power balance, $0.95 \le v_b \le 1.05$ pu, where v_b is bus-voltage magnitude, and

$$\frac{\max_{e\in\{\mathrm{fr,to}\}} |S_{\ell,e}|}{\bar{S}_\ell^{\mathrm{syn}}} \le 1 \quad \text{for every monitored branch } \ell. \tag{4}$$

The branch test uses apparent power in megavolt-amperes (MVA) at both ends, independently of rooted flow orientation; $\bar{S}_\ell^{\mathrm{syn}}$ is the assigned rating of branch ℓ. Supplemental section S8 gives the rating construction, calibration, and AC verification details.

Completed-interval replay and feedback

With unused-award tolerance $\epsilon_x > 0$, let $\mathcal{N}_{\mathrm{ua}}(t) = \{i : x_i(t) - y_i(t) > \epsilon_x\}$. After the interval ends, ECF caps only this group's admitted requests at completed delivery:

$$r_i^{\mathrm{rpl}}(t) = \begin{cases} \min\{r_i^{\mathrm{adm}}(t), y_i(t)\}, & i \in \mathcal{N}_{\mathrm{ua}}(t), \\ r_i^{\mathrm{adm}}(t), & i \notin \mathcal{N}_{\mathrm{ua}}(t). \end{cases} \tag{5}$$

The same allocation rule and network data give replay access $\mathbf{x}^{\mathrm{rpl}}$. Available export then bounds replay delivery:

$$y_i^{\mathrm{rpl}} = \min\{x_i^{\mathrm{rpl}}, a_i^{\mathrm{tel}}\}.$$

The replay recalculates awards, so a member of the unused-award group can receive less access than its completed delivery. The total-delivery condition below subtracts any such loss from recovery elsewhere. Completed awards and delivery stay unchanged. Outside the unused-award group, the test separately sums positive changes in access and delivery. Across all participants, it computes the signed change in total delivery. An interval qualifies only if each quantity exceeds τ_{rpl} and the replay passes the applicable nonlinear AC checks. Supplemental section S3 gives the expressions. The joint request change is tested as a group; the result does not identify each member's contribution.

With exact completed delivery and a validated record-error bound $|a_i^{\mathrm{tel}} - a_i| \le \delta_i$, the conservative replay uses

$$\max\{y_i, a_i^{\mathrm{tel}} - \delta_i\}$$

as a lower bound on available export. Supplemental section S3 proves the resulting lower bound on recovered delivery. Physical checks remain separate. With exact availability and the same recovery threshold, positive aggregate delivery recovery implies positive outside delivery and access recovery. The three quantities trace how released access becomes delivery and are all evaluated for perturbed records. In the exact-record model they are nested conditions, not independent tests.

Let $g(t) = 1$ when all replay conditions hold and zero otherwise. The target and next-interval ceiling are

$$\psi_i^{\mathrm{tar}}(t) = \begin{cases} y_i(t), & g(t) = 1 \text{ and } i \in \mathcal{N}_{\mathrm{ua}}(t), \\ C_i, & \text{otherwise}, \end{cases}$$
$$\psi_i(t+1) = (1 - \beta_{\mathrm{ECF}})\psi_i(t) + \beta_{\mathrm{ECF}}\psi_i^{\mathrm{tar}}(t), \qquad 0 \le \beta_{\mathrm{ECF}} \le 1. \tag{6}$$

For positive gain, a qualifying interval lowers the unused-award group's ceilings toward completed delivery. Other ceilings recover toward connection bounds. Consecutive nonqualifying intervals give geometric recovery for positive gain; a larger raw request cannot bypass a binding ceiling.

The controls distinguish advance incentives from post-delivery updates. The shortfall-cost benchmark changes request choice and has no feedback state. Mismatch-only feedback smooths the participant's delivery-to-award ratio without replay. To remove replay without also changing the update formula, a separate control applies ECF's target and gain after every unused award.

To test whether a similar update frequency reproduces ECF's advantage, 12 fixed replay-free schedules select unused-award intervals with probability $483/3091$, matching ECF's original update fraction among the 3,091 unused-award intervals, not among all 3,840 half-hours. This probability is calibrated retrospectively to make the update frequency comparable to ECF's. Each schedule evolves its own awards, delivery and ceilings, so update magnitudes and durations can also differ. Schedules are averaged within each date. Supplemental section S11.3 gives the selection procedure and all outcomes.

The equal-total control matches ECF's total request reduction in each interval but spreads it proportionally across connections. If $r_i^{\mathrm{cap}} = \min\{r_i, C_i\}$ and $\Delta R_{\mathrm{ECF}} = \sum_i (r_i^{\mathrm{cap}} - r_i^{\mathrm{adm,ECF}})$, the control admits

$$r_i^{\mathrm{adm,uni}} = r_i^{\mathrm{cap}} \left(1 - \frac{\Delta R_{\mathrm{ECF}}}{\sum_j r_j^{\mathrm{cap}}}\right) \tag{7}$$

when the denominator is positive, and retains the capacity-limited request otherwise. This isolates where requests are reduced. With total authorization $X = \sum_i x_i$, total delivery $Y = \sum_i y_i$ and unused authorization $H = \sum_i (x_i - y_i)$, the accounting identity $Y = X - H$ yields

$$Y^{\mathrm{ECF}} - Y^{\mathrm{uni}} = (X^{\mathrm{ECF}} - X^{\mathrm{uni}}) + (H^{\mathrm{uni}} - H^{\mathrm{ECF}}). \tag{8}$$

Summing over intervals, with each power multiplied by its interval duration, gives the corresponding energy balance. Either component can be positive or negative. Supplemental section S4 derives conditions under which redistribution increases delivery.

Table 1 summarizes the controls and the comparison each supports. Their update formulas are specified in supplemental sections S3–S4 and S11.

Controlled experiments

A five-node conditional available-export tree uses standardized support $(-2, -1, 0, 1, 2)$ with probabilities $(1, 4, 6, 4, 1)/16$ and scale σ. The participant's forecast anchors both its request and the later node realization. The static grid spans 30 locations, five rules, Q0 and Q95, five uncertainty levels and nine shortfall-cost levels, giving 13,500 request-choice cases. The selected payoff-maximizing multiplier is held fixed within a 16-interval trajectory. Three temporal profiles have identical individual node counts but differ in ordering: clustered, intermittent and shock–recovery. Participant-specific phase offsets also change the simultaneous combinations of nodes across connections.

Changed requests come from one participant, five electrically nearby participants, or five dispersed participants. In the five-participant cases, each uses its own best-response multiplier and all five requests change simultaneously. The 270 base settings, each evaluated under six mechanisms, give 1,620 trajectories and 25,920 interval comparisons. Table S3 gives every factor, numerical tolerance and parameter level. The post-event record is exact simulated availability in this controlled study.

Australian public-data simulation

The public-data simulation combines a published unbalanced 11-kV feeder with 1,255 service connections and its LV sections[27] with half-hour PV and consumption profiles from 300 Ausgrid solar homes[28]. These sources have no common customer identifier. The primary deterministic profile-to-service mapping is fixed before assessment, and separate placements test its sensitivity. The forward-uptake scenario assigns 418 connections export participation and scales PV to an approximately 9.9-kWp mean[29]. Requests and awards represent this constructed scenario, not historical operating envelopes.

The day-to-day forecast increment is estimated from 2010–2011 data; 2011–2012 data select fixed feeder settings; 2012–2013 provide the assessment chronology. A day is eligible only when mapped profiles are complete on that day and the previous day. Excluding missing and daylight-saving-transition records leaves 80 network days, or 3,840 half-hours per control.

For mapped profile $\iota(i)$, source gross PV generation GG, general consumption GC and controlled load CL are first converted to interval-average MW. Applying the scenario multipliers gives

$$a_i^{\mathrm{prof}}(d,\tau) = \left[\lambda_{\mathrm{PV}}\mathrm{GG}_{\iota(i)}(d,\tau) - \lambda_L(\mathrm{GC}_{\iota(i)}(d,\tau) + \mathrm{CL}_{\iota(i)}(d,\tau))\right]^+ . \tag{9}$$

Modeled delivery under control c is $y_i^c = \min\{x_i^c, a_i^{\mathrm{prof}}\}$; the unperturbed profile proxy also supplies the primary replay record. Raw requests use the previous day's same-half-hour proxy plus a positive 90th-percentile increment from same-half-hour day-to-day changes. Contemporaneous household demand is fixed as an exogenous input to this retrospective allocation; load-forecast error is not modeled.

For AC delivery calculations, household consumption is first offset by local PV; the remaining net import minus delivered net export is the active-power load. The authorization screen instead sets participant net injection equal to the full award, without using realized PV to relax the constraint. Household reactive demand is retained in both screens. Supplemental section S10.4 gives the connection-level equations and mapping.

The primary feeder settings use a 0.30 load multiplier and a 1.07-pu allocation-stage upper-voltage limit, below the 1.10-pu physical limit. For each half-hour, a zero-anchored feasible-factor search scales admitted requests under nonlinear unbalanced AC power flow. An independent

factor-grid check of all 19,200 primary allocation calls found no feasible sampled factor above the returned one; supplemental section S10.4 states the search resolution and its limits. Load-only, authorized-export and modeled-delivery states are screened separately for convergence, customer voltage and native transformer ratings. Native line ampacities are unavailable, so the public-study screen excludes line thermal limits. The 2 W replay threshold suppresses numerical near-zero differences; record-error bounds are specified separately.

Each primary day begins with ceilings at C_i and compares no feedback, previous-day persistence, ECF at $\beta_{\mathrm{ECF}} = 0.5$, and equal-total reduction. The additional comparisons vary gain, replay qualification, cross-day memory, record perturbations and error bounds, profile placement, PV capacity, participant adaptation and published LV networks while retaining the documented fixed inputs for each matched comparison. Source-terminal power from the same AC solves accounts for upstream exchange.

The request quantile and feeder settings use the earlier source years, with initial exploratory runs also informing the operating setting. All reported 80-day comparisons use the resulting common setting. Supplemental sections S10.3–S10.4 give the construction sequence and full model settings.

Statistical analysis

The primary comparison unit in the Australian study is one paired network day. A day contains 48 half-hours on the same network and profile mapping; participants, half-hours and nested LV sections are not treated as independent network samples. Energy differences are computed within each day before averaging. We report ordinary paired-day percentile intervals and date-dependent wild-bootstrap intervals, which allow short-range correlation between nearby eligible dates. Both procedures use 10,000 replicates. For request adaptation, three seeds are averaged within each date. For the random-update comparison, 12 schedules are averaged within each date. Intervals are pointwise, without adjustment for multiple comparisons. Supplemental section S10.3 gives the calendar covariance, eligibility rules and numerical settings.

SUPPLEMENTAL INFORMATION

Document S1 contains supplemental methods, derivations, Figures S1–S6, and supplemental tables. Data S1 contains the figure source-data workbooks and their figure-to-worksheet map.

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AUTHOR CONTRIBUTIONS

Junjie H.: conceptualization, methodology, software, formal analysis, investigation, visualization, data curation, and writing—original draft. T.Y.: conceptualization, supervision, funding acquisition, and writing—review and editing. Z.P., Y.W., and W.X.: methodology, validation, and

writing—review and editing. R.Z.: validation, data curation, and writing—review and editing. Jie H.: validation and writing—review and editing.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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